

Practical Assessment 02

Subject: Computational Financial Modeling

1. Assessment Objectives

The objective of this assessment is to apply data science principles to financial mathematics. Students must demonstrate the ability to translate mathematical models into precise computational algorithms. Key focus areas include:

- **Algorithmic Implementation:** Developing functions that accurately model financial datasets.
- **Mathematical Translation:** Mapping complex formulas into optimized Python logic.
- **Numerical Precision:** Managing floating-point data types to ensure high-fidelity financial reporting (ZAR).
- **Model Validation:** Verifying algorithmic accuracy against known historical reference points.

2. Technical Requirements

- **Environment:** Python 3.x
- **Data Consistency:** All financial outputs must be formatted to two decimal places to maintain currency integrity.
- **Validation:** Calculated results must align exactly with the provided Reference Outputs.

3. Implementation Tasks

ID	Requirement & Mathematical Model	Implementation & Reference Output
01	Simple Interest: Principal of R4,500 at 7% p.a. for 5 years. <i>Model:</i> $I = P \times r \times t$	<code>calculate_si(4500, 0.07, 5)</code> Reference: R1 575.00
02	Compound Interest: Principal of R12,000 at 6.5% for 8 years. <i>Model:</i> $A = P(1 + i)^n$	<code>compound_ann(12000, 0.065, 8)</code> Reference: R19 860.13
03	Hire Purchase: Cost R22,000, 15% deposit, 11% SI, 3 years. <i>Model:</i> $A = [P_{bal}(1 + rt)] \div 36$	<code>hp_monthly(22000, 0.15, 0.11, 3)</code> Reference: R690.03
04	Inflation Analysis: Present cost R1,550, 5.5% rate, 12 years. <i>Model:</i> $A = P(1 + i)^n$	<code>inflation(1550, 0.055, 12)</code> Reference: R2 946.90
05	Reducing Depreciation: R480,000 at 18% for 6 years. <i>Model:</i> $A = P(1 - i)^n$	<code>depreciate(480000, 0.18, 6)</code> Reference: R145 920.44
06	Quarterly Compounding: R95,000 at 9% for 4 years. <i>Model:</i> $A = P(1 + \frac{r}{4})^{4t}$	<code>comp_q(95000, 0.09, 4)</code> Reference: R135 621.84
07	Loan Accrual: R30,000 at 14% for 1 year. <i>Model:</i> $I = [P(1 + \frac{r}{12})^{12}] - P$	<code>loan_acc(30000, 0.14, 1)</code> Reference: R4 480.25
08	Capital Doubling Time: R15,000 at 12.5% Simple Interest. <i>Model:</i> $t = I / (P \times r)$	<code>double_time(15000, 0.125)</code> Reference: 8.0 Years
09	Effective Annual Rate: 13.2% Nominal monthly to EAR. <i>Model:</i> $i_{eff} = (1 + \frac{i_{nom}}{m})^m - 1$	<code>get_ear(0.132, 12)</code> Reference: 14.03%
10	Fund Growth: R2,500,000 at 15% semi-annually for 10y. <i>Model:</i> $A = P(1 + \frac{r}{2})^{2t}$	<code>zaio_growth(2500000, 0.15, 10)</code> Reference: R10 804 532.18